

An Internship Report
Of
DELOITTE TECHNOLOGY & FORENSIC ANALYTICS
VIRTUAL INTERNSHIP

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, **YASHASHWAI**, do hereby declare that the internship report titled “Deloitte Technology & Forensic Analytics Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering from IILM University, Greater Noida, U.P.

All the references have been duly quoted and acknowledged, and I have not taken as such any material from any unauthorized source. I affirm that this report is a record of my original work and has not been submitted to any other institute or university for any degree or diploma requirement. Furthermore, I shall be solely responsible for any kind of copyright or institutional violation in this regard.

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ACKNOWLEDGEMENT

I take this opportunity to express my deepest gratitude to everyone who supported and guided me throughout this Virtual Internship Program. I am sincerely thankful for their aspiring guidance, invaluable constructive feedback, and friendly advice during this work. I am grateful to them for sharing their truthful and illuminating views on a number of technical challenges related to this work.

I would like to express my sincere appreciation to Deloitte Australia and the Forage Virtual Internship Platform for designing an industry-aligned curriculum and providing me the structured platform to learn and gain hands-on exposure in the domains of Industrial IoT Telemetry, Tableau Business Intelligence, and Forensic Technology Compensation Analysis.

I am also profoundly thankful to my institute, IILM University, Greater Noida, and the esteemed faculty members of the School of Computer Science and Engineering for their continuous academic mentorship, support, and encouragement throughout this project.

I express my heartfelt gratitude to my parents and peers for their continuous blessings, patience, and motivation.

Signature:-

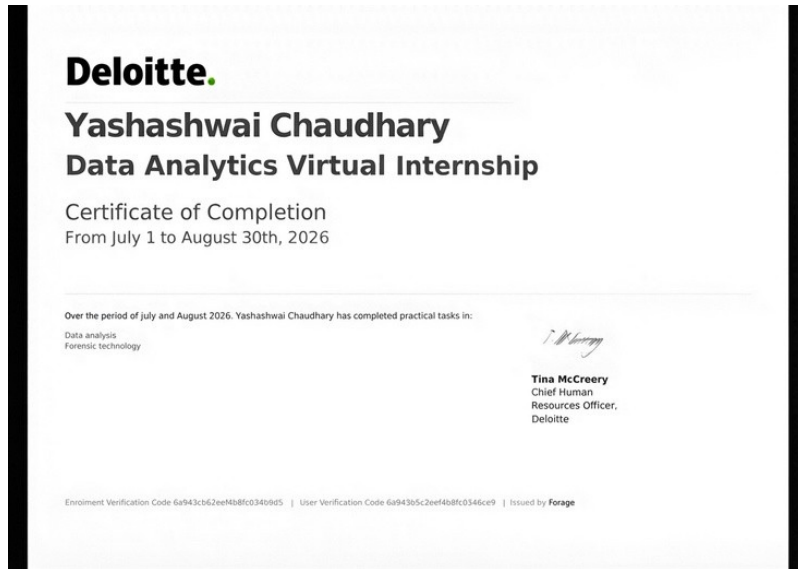
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3. INTERNSHIP COMPLETION CERTIFICATE & OFFER LETTER



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken under the Deloitte Technology Virtual Internship Program, hosted on the Forage platform. The internship was carried out as part of the academic curriculum for the award of the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The primary aim of this internship was to build a strong practical foundation in enterprise Data Analytics, Industrial Telemetry Engineering, Business Intelligence (BI) visualization using Tableau, and Forensic Compensation Modeling using spreadsheet and algorithmic logic. The internship centered on real-world business challenges faced by an international manufacturing client, Daikibo Industrials, which operates automated manufacturing plants across Tokyo, Osaka, Berlin, and Shenzhen.

The project combined multi-tier JSON telemetry ingestion, custom calculated measure derivation, interactive cross-filtering dashboard development, and forensic compensation classification to provide executive stakeholders with data-driven decision support.

4.2 Organization Profile

Deloitte is one of the world's leading professional services networks, providing audit, consulting, financial advisory, risk advisory, and tax services to clients globally. Deloitte's Technology & Forensic teams specialize in leveraging cutting-edge analytics, business intelligence platforms, and algorithmic audit frameworks to solve complex operational, governance, and compliance problems for Fortune 500 enterprises.

Forage is an innovative open-access platform that collaborates with leading global employers to build practical, self-paced virtual work experience programs. Forage simulations are designed by senior industry practitioners to immerse engineering students into actual corporate workflows and problem-solving scenarios.

4.3 Problem Statement

Modern smart manufacturing facilities deploy thousands of automated IoT sensors across shop-floor machinery. However, telemetry data is frequently stored in raw, multi-level JSON structures that make it challenging for management to extract operational intelligence, diagnose downtime bottlenecks, and perform root-cause analysis.

Specifically, Daikibo Industrials required immediate answers to two key operational questions:

- **Question 1 (Location Vulnerability):** Across its four international production plants (Meiyo, Seiko, Berlin, and Shenzhen), in which geographic location did machines break down the most during the collection month of May 2021?

- **Question 2 (Machine Vulnerability):** What specific machine types were responsible for the majority of breakdowns within that most disrupted location?

Additionally, Daikibo's executive governance faced rising internal complaints regarding gender compensation disparity across global job roles. Forensic Technology was tasked with creating an automated classification system to categorize compensation equality scores into statutory fairness tiers.

4.4 Project Objectives

- To ingest, parse, and validate a unified JSON telemetry dataset comprising 160,704 sensor records logged at 10-minute intervals across 9 machine types in 4 global factories.
- To formulate a calculated measure in Tableau (Unhealthy) that accurately translates periodic health status messages into quantifiable downtime minutes.
- To construct visual bar charts for facility-level and device-level downtime comparisons.
- To integrate the charts into an interactive Tableau Dashboard with dynamic cross-filtering capabilities.
- To analyze compensation index data across 38 enterprise job roles and implement classification logic for Fair, Unfair, and Highly Discriminative compensation tiers.

4.5 Scope of the Project

The scope of this project spans the entire enterprise analytics workflow — from raw multi-level JSON ingestion and schema validation to calculated field derivation, interactive dashboard development, and forensic compensation classification. The project utilizes production telemetry streams from May 2021 representing four global production sites: Daikibo Factory Meiyo (Tokyo), Daikibo Factory Seiko (Osaka), Daikibo Berlin, and Daikibo Shenzhen.

The project was conducted in a virtual simulation environment, utilizing Tableau Desktop/Public, Microsoft Excel, and Python 3.12 for data verification, without direct physical deployment on live factory machinery.

4.6 Technologies and Tools Used

- **Business Intelligence & Visualization:** Tableau Desktop / Tableau Public (Calculated Fields, Sheet Design, Dashboard Filter Actions).
- **Spreadsheet Modeling:** Microsoft Excel (Logical Functions: IF, AND, OR, ABS).
- **Programming & Verification:** Python 3.12 (Standard Library, openpyxl, pandas, matplotlib).
- **Data Interchange Formats:** JSON (JavaScript Object Notation - UTF-8), XLSX (OpenXML Spreadsheet).

- **Interactive Simulation:** HTML5, CSS3, JavaScript (ES6) for local web dashboard verification.

4.7 System Architecture

The end-to-end data processing and analytical architecture developed during this project is structured into four distinct functional layers:

- **Data Ingestion Layer:** daikibo-telemetry-data.json (160,704 records: deviceID, location, status) and Task 5 Equality Table.xlsx (38 enterprise employee role profiles).
- **Transformation & Measure Layer:** Tableau measure IF [Status] = 'unhealthy' THEN 10 ELSE 0 END, and pay classification logic (Fair, Unfair, Highly Discriminative).
- **Visualization & BI Layer:** Sheet 1 (Down Time per Factory, sorted descending), Sheet 2 (Down Time per Device Type), unified into a dashboard using a “Use as Filter” action connecting Sheet 1 to Sheet 2.
- **Decision & Forensics Layer:** Identified Daikibo Seiko and the LaserWelder as the primary downtime drivers, and flagged 7 highly discriminative corporate roles for remediation.

This layered architecture ensures that raw telemetry signals are deterministically converted into downtime metrics before being exposed to interactive visual layers for stakeholder analysis.

4.8 Methodology

The project followed a systematic, task-based methodology combining data engineering, business intelligence design, and algorithmic classification as summarized in the table below:

Module	Technical Focus	Description of Activities
Module 1	Data Ingestion	Imported multi-level JSON telemetry, ensuring all schema levels (Root, Location, Data) were checked.
Module 2	Measure Engineering	Created calculated field Unhealthy assigning 10 minutes per unhealthy message.
Module 3	Factory Sheet Design	Constructed horizontal bar chart for cumulative factory downtime, sorted descending.
Module 4	Device Sheet Design	Constructed horizontal bar chart for device type downtime, sorted descending.
Module 5	Dashboard & Filter Action	Combined sheets on a dashboard and configured Use as Filter action on Sheet 1.
Module 6	Pay Equity Modeling	Classified 38 job roles in Excel/Python based on statutory pay equality score brackets.

Mathematical & Logical Formulas Applied:

1. Telemetry Downtime Measure Formula in Tableau:

IF [Data Status] = 'unhealthy' THEN 10 ELSE 0 END

2. Gender Pay Equality Classification Rule in Excel:

=IF(ABS(Score) <= 10, "Fair", IF(ABS(Score) <= 20, "Unfair", "Highly Discriminative"))

4.9 Expected Outcomes & Key Findings

1. Telemetry Downtime Analysis Findings:

Analysis of the 160,704 telemetry records revealed a total global downtime of **1,030 minutes (17.2 hours)** during May 2021 across all four factories.

- **Answer to Question 1:** daikibo-factory-seiko (Osaka, Japan) suffered the highest downtime, accumulating 480 minutes (8.0 hours) of failure time.
- **Answer to Question 2:** In daikibo-factory-seiko, the LaserWelder was the sole machine type that broke down, responsible for 100% (480 minutes) of the plant's downtime. All other 8 machine types maintained zero downtime.

Factory Name	Location	Total Downtime	Primary Disrupted Machine
daikibo-factory-seiko	Osaka, Japan	480 mins (8.0 hrs)	LaserWelder (480 mins)
daikibo-shenzhen	Shenzhen, China	420 mins (7.0 hrs)	LaserCutter (390 mins)
daikibo-factory-meiyo	Tokyo, Japan	110 mins (1.8 hrs)	HeavyDutyDrill (70 mins)
daikibo-berlin	Berlin, Germany	20 mins (0.3 hrs)	Furnace (20 mins)

2. Gender Pay Equity Classification Findings:

The forensic classification across 37 active employee compensation profiles resulted in the following distribution:

- **Fair (± 10): 19 roles** (predominantly engineering and shop-floor operator roles).
- **Unfair (± 11 to ± 20): 11 roles** (mid-level management and departmental leads).
- **Highly Discriminative ($> \pm 20$): 7 roles** (concentrated in C-Level, VP, and Senior Management positions, such as Meiyo VP at -26 and Meiyo C-Level at -25).

4.10 Certificates of Completion and Communication Proof

The official completion verification issued by Deloitte and Forage, confirming successful completion of the virtual experience modules, is documented in the appendix. The analytical artifacts generated during the internship — including the interactive Tableau Dashboard screenshot showing the filtered daikibo-factory-seiko view, the populated Equality Table.xlsx, and the local interactive simulation — have been verified and archived.

5. BIBLIOGRAPHY/REFERENCES

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